Do Tail-Risk Models Add Value Beyond Volatility and Correlation?

Evidence from Multi-Asset Markets

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October 2026

Abstract

Tail-risk methods (copula tail dependence, expected shortfall budgeting) are widely promoted in sell-side research as improvements over volatility- and correlation-based risk management. This note asks whether they carry information that volatility and correlation do not. Two tests are run on Chinese and global asset classes. *Test 1* compares tail-dependence statistics with a Gaussian-copula null that matches the observed correlation matrix in every estimation block. Single-parameter Clayton/Gumbel coefficients, the quantities used in practice, are statistically indistinguishable from this null: their rank correlation with the ordinary correlation is 0.94 in the data and 0.94 under the null. Models with a separate tail parameter do detect dependence beyond correlation in a six-asset weekly panel (31 of 105 Student-t versus Gaussian tests reject at 5%, against 5.7 expected under the matched null), concentrated in risky-asset pairs and in the lower tail; between Chinese stocks and government bonds the evidence is weak and has no downside asymmetry. *Test 2* decomposes tail risk parity into its two ingredients: replacing volatility by expected shortfall, and conditioning on GARCH volatility forecasts. Replacing volatility by ES adds 0.00–0.04 to the excess Sharpe ratio, never significantly. Conditioning adds about 0.11 before costs, but raises annual one-way turnover from 15% to 95%, and the gain is gone at 25 basis points per unit traded. A preliminary section shows that the Acerbi–Szekely Z_2 threshold of -0.70, calibrated for 250 observations, is uninformative in long backtests, and replaces it with a self-normalised test with a wild-bootstrap p-value. Within the markets and sample studied, the incremental value of these tail-risk methods over tracking current volatility and correlation is small.

1 Introduction

Volatility and linear correlation describe a multivariate return distribution only up to its second moments. Tail-risk methods promise more: copula tail-dependence coefficients are meant to capture joint crashes that correlation misses (Embrechts et al., 2002), and expected shortfall (ES) is a coherent measure of the loss beyond the value-at-risk (VaR). Chinese sell-side reports have turned these ideas into trading rules, for example a stock–bond "double-kill" warning triggered by rolling Clayton and Gumbel tail coefficients, hierarchical allocation on lower-tail distances, and tail risk parity (Alankar et al., 2013). The question this note asks is narrow: *given current volatility and correlation, do these tail-risk statistics add information?*

Two difficulties make the question harder than it looks. First, the natural benchmark is not independence. A single-parameter Archimedean copula fitted to positively correlated data returns a positive tail coefficient even when the true copula is Gaussian and has no tail dependence. The

*Research note, not peer reviewed. Code and data-retrieval notebooks: `https://github.com/lshangro-png/tail-risk-cn`. All results can be regenerated from free public data sources.

null has to hold the correlation fixed. Second, an allocation built on ES may differ from one built on volatility for two separate reasons: the risk measure changes, and the scenarios used to estimate it are often conditioned on current volatility. These two effects have to be separated, and the comparison has to be made net of the turnover each one generates.

The note makes three contributions. (i) It runs a *correlation-matched* Gaussian-copula null for a panel of tail-dependence statistics, which handles both the correlation confound and the dependence between tests that share an asset. (ii) It decomposes tail risk parity into a 2×2 design (risk measure $\times$ conditioning) and evaluates it under two explicit execution rules with realised turnover and transaction costs. (iii) It shows that the fixed Z_2 threshold commonly used to backtest ES is not calibrated for long samples and gives a simple replacement.

The underlying replications of sell-side reports, with their code, are in the accompanying repository. This note summarises the two strongest lines of evidence from that work.

2 Data

All data come from free public sources: the ChinaBond website (bond wealth indices), Sina, Tencent and Eastmoney via AkShare (Chinese indices and ETFs), FRED (exchange rates, Brent, volatility indices) and the Kenneth French library. Prices are stored in a local SQLite database and every result is recomputed from it.

- **Single-asset backtests (Section 3).** Daily returns of the SSE 50, CSI 300 and CSI 500 indices, the ChinaBond composite wealth index, gold (GLD $\times$ USD/CNY), the S&P 500 and Brent, from 1987–2006 (depending on the series) to October 2026.
- **Stock–bond pair (Section 4).** CSI 300 and ChinaBond composite wealth index, daily, 2003–2026. Both close at the same time in Beijing.
- **Multi-asset panel (Section 4) and allocation (Section 5).** CSI 300, CSI 500, S&P 500, ChinaBond government-bond total wealth index (CGB), gold in CNY and Brent. Chinese and Western markets close at different times, which biases same-day dependence towards zero, so the dependence tests use *weekly* (Friday-to-Friday) returns. The allocation backtest is monthly and uses daily returns from 2005 to October 2026 (5,008 common trading days).
- **ETF panel.** Ten Chinese-listed ETFs (domestic equity styles, Nasdaq, Hang Seng, gold, commodities), 2016–2026, used for the lower-tail-distance comparison.

3 Preliminary: marginal tails and a calibrated ES backtest

Before testing dependence, it is worth checking what matters for one asset at a time. Six one-day-ahead VaR/ES forecasters are run on rolling 1,000-day windows: historical simulation, Gaussian, Student-t, GPD peaks-over-threshold, GJR-GARCH(1,1) with filtered historical simulation (FHS), and GJR-GARCH with a GPD tail on the standardised residuals (GARCH-EVT; McNeil and Frey, 2000; Glosten et al., 1993). The 99% VaR is evaluated with the Kupiec (1995) unconditional-coverage test and the Christoffersen (1998) independence test, and the 97.5% ES with the Z_2 statistic of Acerbi and Szekely (2014).

Table 1: Rejection rates of the 97.5% ES backtest in 1,000 simulations of a GARCH(1,1) with standardised t_5 innovations. "Rule": $Z_2 < -0.70$. "Test": one-sided asymptotic test on S at 5%.

	$T = 250$		$T = 5000$	
Forecast	Rule	Test	Rule	Test
Correct model	5.3%	1.7%	0.0%	4.3%
Gaussian tail, correct volatility	15.7%	4.5%	0.0%	61.5%
VaR and ES scaled by 0.85	44.5%	25.3%	37.0%	100.0%

The Z_2 test needs a null distribution, not a fixed threshold. Let X_t be the return, VaR_t and ES_t the forecasts at level α (both positive numbers), and $I_t = \mathbf{1}\{X_t < -\mathrm{VaR}_t\}$. Acerbi and Szekely define

$$Z_2 = \frac{1}{T\alpha} \sum_{t=1}^{T} \frac{X_t I_t}{\mathrm{ES}_t} + 1. \tag{1}$$

Write $Y_t = X_t I_t / \mathrm{ES}_t + \alpha$, so $Z_2 = \sum_t Y_t / (T\alpha)$. If both forecasts are correct, $\mathbb{E}_{t-1}[X_t I_t] = -\alpha\, \mathrm{ES}_t$, hence $\mathbb{E}_{t-1}[Y_t] = 0$: $\{Y_t\}$ is a martingale-difference sequence, whatever the volatility dynamics. Acerbi and Szekely report -0.70 as an approximate 5% critical value, obtained by simulation with $T = 250$. The null standard deviation of Z_2 shrinks like $1/\sqrt{T\alpha}$, so in a backtest of $T \approx 5{,}000$ days a fixed -0.70 corresponds to a far smaller test size and much lower power. I therefore use the self-normalised statistic

$$S = \frac{\sum_t Y_t}{\sqrt{\sum_t Y_t^2}} \xrightarrow{d} N(0,1) \quad \text{under } H_0, \tag{2}$$

which follows from the martingale central limit theorem (Hall and Heyde, 1980). A finite-sample p-value is obtained by a wild bootstrap: $S^* = \sum_t Y_t \varepsilon_t / \sqrt{\sum_t Y_t^2}$ with independent Rademacher ε_t (Wu, 1986; Mammen, 1993). The test is one-sided: a significantly negative S means the ES is too small or VaR violations are too frequent. A related cumulative-violation approach is Du and Escanciano (2017).

Monte Carlo check. Table 1 simulates a GARCH(1,1) with standardised t_5 innovations ($\omega = 0.02$, $\alpha = 0.08$, $\beta = 0.90$, returns in percent) and evaluates three forecasts of the 97.5% ES: the true conditional VaR and ES; a Gaussian tail with the true volatility, which understates ES by 14% while VaR is only 1.6% too low; and the true forecasts scaled by 0.85. With $T = 250$ the rule $Z_2 < -0.70$ rejects the correct model in 5.3% of samples, as designed. With $T = 5{,}000$ it never rejects the correct model, never detects the Gaussian tail, and detects the 15% scaling in only 37% of samples. The test on S has size 4.3% at $T = 5{,}000$ and power of 62% and 100% against the two misspecifications. At $T = 250$ the asymptotic test is conservative (size 1.7%; a year contains only about six 97.5% violations), and the wild-bootstrap p-value should be used. In the long backtests below, asymptotic and bootstrap p-values agree to within 0.03.

Results. Table 2 reports, for each method, the number of the seven assets for which each test does not reject at 5%. With the fixed threshold, the Gaussian model "passes" the ES backtest for six of seven assets, although its realised tail losses exceed the forecast ES by 27% (median). With the calibrated test it is rejected for all seven. Methods without volatility updating (historical simulation, Student-t, GPD) have roughly correct unconditional coverage, but their violations cluster: the probability of a violation on the day after a violation is 4–15% instead of 1%, and the

Table 2: One-day-ahead VaR/ES backtests on seven assets (rolling 1,000-day windows). Columns 3–6 count the assets (of 7) for which the test does not reject at 5%. "Old rule": $Z_2 > -0.70$. "Calibrated": one-sided wild-bootstrap p-value of the self-normalised statistic S. Last column: median over assets of mean realised loss over mean forecast ES on 97.5% VaR violation days.

Method	99% VaR hit rate	Kupiec	Christoffersen	Z_2 old rule	Z_2 calibrated	Loss/ES
Historical simulation	1.22%	5/7	1/7	7/7	4/7	1.03
Gaussian	1.83%	0/7	0/7	6/7	0/7	1.27
Student-t	1.13%	3/7	1/7	7/7	2/7	1.00
GPD (POT)	1.12%	5/7	1/7	7/7	4/7	1.03
GARCH-FHS	1.07%	7/7	5/7	7/7	7/7	1.03
GARCH-EVT	0.97%	7/7	6/7	7/7	7/7	1.03

Christoffersen test rejects for six of seven assets. Their ES is rejected for three to five assets. Both GARCH filters pass all three tests for nearly every asset; adding a GPD tail to the filtered residuals (GARCH-EVT) improves clustering slightly over FHS (6/7 versus 5/7).

The lesson for what follows is that, at the level of a single asset, the information that matters is the current volatility. Tail-shape modelling on top of a GARCH filter (EVT versus FHS) changes little.

4 Test 1: tail dependence beyond correlation

4.1 Why single-parameter tail coefficients mostly measure correlation

The Clayton copula $C_\theta(u, v) = (u^{-\theta} + v^{-\theta} - 1)^{-1/\theta}$ has lower tail dependence $\lambda_L = 2^{-1/\theta}$; the Gumbel copula has $\lambda_U = 2 - 2^{1/\theta}$. Each family has one parameter, which governs both the overall strength of positive dependence and the tail. When either is fitted to data whose true copula is Gaussian with correlation $\rho > 0$, maximum likelihood moves θ away from the independence boundary to match the overall dependence, and $\hat{\lambda}$ becomes positive although the true tail dependence is zero (Joe, 1997). The appropriate null is therefore a Gaussian copula with the same correlation, not independence.

On the CSI 300–ChinaBond pair, with rolling 60-day windows as in the CICC (2021) warning rule, this matters a great deal. In a Gaussian copula with $\rho = 0.3$, 94% of 60-day windows already give $\hat{\lambda}_L > 0$ under the point-estimate rule. The 60-day correlation alone separates days with $\hat{\lambda}_L > 0$ from days with $\hat{\lambda}_L = 0$ with an AUC of 0.95. Against a Gaussian null that matches each window's normal-score correlation, the observed frequency of $\hat{\lambda}_L > 0$ is 33.4% versus 34.5% expected, and the frequency of the CICC trigger ($\hat{\lambda}_L > 0$, $\hat{\lambda}_U = 0$) is 4.8% versus 4.7%. The signal is what time-varying correlation produces on its own. In the ten-ETF panel, the rank correlation between pairwise Clayton $\hat{\lambda}_L$ (250-day windows) and the ordinary correlation averages 0.98 across rebalancing dates, and a portfolio-ES aggregation formula using $\hat{\lambda}_L$ is no more accurate than one using ρ (median absolute error 2.2% versus 2.0%).

4.2 Models with a separate tail parameter

To ask whether there is tail dependence *beyond* correlation, the model needs a parameter for the tail that is separate from the one for overall dependence. Two are used.

Student-t versus Gaussian copula. The t copula has correlation ρ and degrees of freedom ν;

its tail dependence $\lambda = 2\,t_{\nu+1}\Big(-\sqrt{(\nu+1)(1-\rho)/(1+\rho)}\Big)$ vanishes as $\nu \to \infty$, where it becomes Gaussian (Demarta and McNeil, 2005). With the same ρ, a finite ν means extremes occur together more often than the Gaussian copula implies. The likelihood-ratio statistic for $H_0 : \nu = \infty$ lies on the boundary of the parameter space, so its null distribution is $\frac{1}{2}\chi^2_0 + \frac{1}{2}\chi^2_1$ and $p = \frac{1}{2}P(\chi^2_1 \geq \mathrm{LR})$.

Symmetrised Joe–Clayton (SJC) copula (Patton, 2006). Its two parameters are the lower and upper tail-dependence coefficients τ_L, τ_U, so $H_0 : \tau_L = \tau_U$ is a one-restriction likelihood-ratio test of tail asymmetry.

For the stock–bond pair, the t test is run in each calendar year (non-overlapping windows, so rejections can be counted with a binomial test). It rejects Gaussianity at 5% in 4 of 23 years against 1.15 expected (binomial $p = 0.026$), and in 7 of 22 years after scaling returns by trailing 60-day volatility ($p = 10^{-4}$). The implied tail dependence is small (median 0.2% over rolling 250-day windows). The direction is symmetric: both assets fall into their worst 10% on the same day 57 times and into their best 10% 55 times, against 47.5 expected under the Gaussian copula. The SJC test finds no downside asymmetry (pooled $p = 0.79$ over the five years with any measurable tail dependence). Between Chinese stocks and government bonds, the dependence beyond correlation is weak and does not favour joint crashes.

4.3 A multi-asset panel with a correlation-matched null

The stock–bond pair is one pair. I extend the tests to all 15 pairs of the six allocation assets, in seven non-overlapping three-year blocks of weekly returns (2005–2025, 139–153 weeks per block), giving 105 pair-blocks. Within a block, ranks are converted to pseudo-observations and, for each pair, I record the t versus Gaussian p-value, the t-implied λ, the SJC estimates and asymmetry p-value, the Clayton $\hat{\lambda}_L$, and the number of weeks in which both assets fall into their lowest (highest) decile.

Null. Tests that share an asset are dependent, so the number of rejections is not binomial, and the t test is mildly oversized for strongly correlated pairs (8% at $\rho = 0.6$ with 156 observations). Both problems are handled by simulating the whole panel under the null. In each block, the null is a six-dimensional Gaussian copula whose correlation matrix equals the block's normal-score correlation matrix. It therefore reproduces the cross-sectional correlation structure and its variation across blocks, and has no tail dependence. Data of the same length are simulated, every statistic is recomputed exactly as for the observed data, and the procedure is repeated 200 times. Each summary statistic is compared with its simulated null distribution, with one-sided p-values $(1 + \#\{s^* \geq s\})/(1 + 200)$.

Table 3 and Figure 1 give two clear answers.

The Clayton coefficient carries no information beyond correlation. Its rank correlation with ρ is 0.94 in the data and 0.94 under the null; the mean $\hat{\lambda}_L$ is only slightly higher than under the null (0.096 versus 0.082). Whatever tail dependence exists, the statistic used by the sell-side rules is almost entirely a monotone function of correlation.

There is tail dependence beyond correlation, and it is stronger in the lower tail. The t test rejects in 31 of 105 pair-blocks against 5.7 expected under the matched null ($p < 0.01$). Joint lower-decile weeks exceed joint upper-decile weeks by 51, against 4 under the null ($p = 0.015$), and SJC estimates have $\tau_L > \tau_U$ on average. Scaling returns by their trailing 26-week volatility, which removes common volatility regimes, leaves these results unchanged or slightly stronger. The t-test rejections are spread across many pairs, including some involving government bonds. The lower-tail excess, however, comes from pairs of risky assets: S&P 500–Brent (29 joint lower-decile

Table 3: Multi-asset panel (6 assets, 15 pairs, 7 non-overlapping three-year blocks of weekly returns, 105 pair-blocks) against the correlation-matched Gaussian-copula null (200 simulations). p: one-sided simulated p-value (large values of the statistic). Volatility-scaled: returns divided by trailing 26-week volatility.

	Raw weekly returns			Volatility-scaled		
Statistic	Obs.	Null mean [5%, 95%]	p	Obs.	Null mean	p
# t vs Gaussian rejections (5%)	31	6 [1, 10]	0.005	31	6	0.005
Mean t-implied λ	0.063	0.018 [0.010, 0.028]	0.005	0.064	0.018	0.005
Joint lower-decile weeks	302	235 [212, 257]	0.005	299	226	0.005
Joint upper-decile weeks	251	231 [207, 257]	0.104	226	224	0.438
Lower − upper	51	4 [-25, 32]	0.015	73	3	0.005
Mean SJC $\hat{\tau}_L - \hat{\tau}_U$	0.049	0.000 [-0.016, 0.018]	0.005	0.055	0.001	0.005
# SJC asymmetry rejections (5%)	8	2 [0, 5]	0.005	7	2	0.010
Mean Clayton $\hat{\lambda}_L$	0.096	0.082 [0.073, 0.093]	0.015	0.094	0.079	0.020
Spearman($\hat{\lambda}_L, \rho$)	0.94	0.94 [0.92, 0.96]	0.512	0.94	0.94	0.413

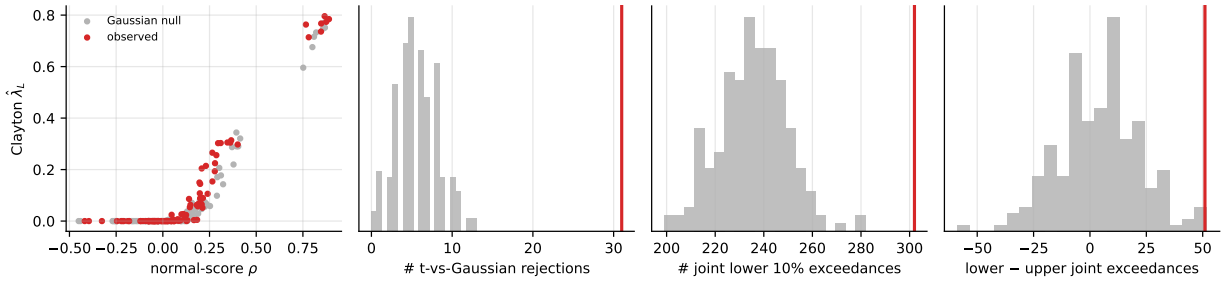


Figure 1: Multi-asset panel, raw weekly returns. Left: Clayton $\hat{\lambda}_L$ against the normal-score correlation for the 105 pair-blocks (red) and for one simulation of the matched Gaussian null (grey). Other panels: null distributions (grey) of three summary statistics and the observed value (red line).

weeks against 18 upper), gold–Brent (26 against 13) and CSI 300–S&P 500 (26 against 16). Pairs involving government bonds show no lower-tail excess.

So the answer to Test 1 depends on the statistic. Tail dependence beyond correlation is statistically detectable in weekly data, with a lower-tail excess among risky assets. It is not visible in the single-parameter coefficients used in practice, and between Chinese equities and bonds, the pair that most stock–bond warning rules target, it is small and symmetric.

5 Test 2: does ES risk budgeting add value?

5.1 Design

Risk parity chooses long-only weights w so that each asset's contribution to a risk measure R is equal. For a positively homogeneous R, the equal-risk-contribution portfolio is $w = y^*/\mathbf{1}'y^*$, with

$$y^* = \arg\min_{y>0} \; R(y) - \sum_{i=1}^{n} b_i \log y_i, \qquad b_i = 1/n, \tag{3}$$

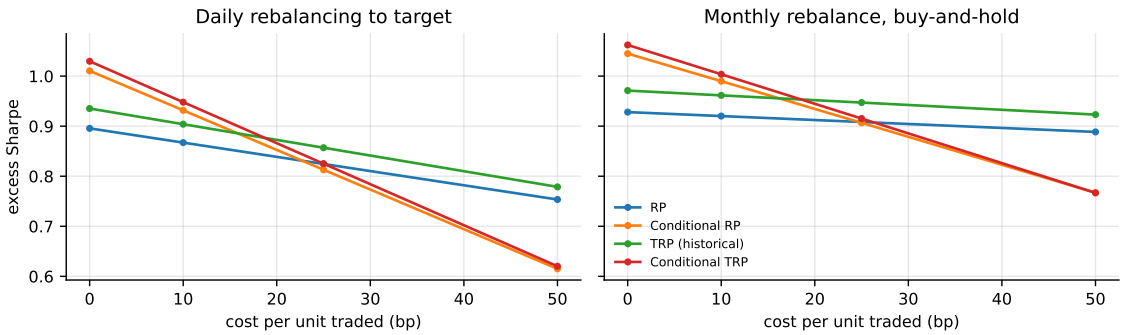


Figure 2: Excess Sharpe ratio against the transaction cost per unit traded, 2007-07 to 2026-10.

a convex problem whenever R is convex (Spinu, 2013; Maillard et al., 2010). For volatility, $R(y) = \frac{1}{2}y'\Sigma y$ (the quadratic form, which yields the same solution as $\sqrt{y'\Sigma y}$). For ES at level q on scenarios $x_1, \ldots, x_T$, the Rockafellar and Uryasev (2000) representation gives a linear program inside the objective:

$$R(y) = \min_{a,\, u \geq 0} \; a + \frac{1}{(1-q)T} \sum_{t=1}^{T} u_t \quad \text{s.t.} \quad u_t \geq -x_t' y - a. \tag{4}$$

This formulation keeps all ES contributions positive and equal at the optimum even when a hedge asset has a negative contribution at equal weights, a case where fixed-point iterations fail (Mausser and Romanko, 2018).

Four strategies form a 2×2 design:

	Unconditional scenarios	Conditional (FHS) scenarios
Volatility budget	RP	Conditional RP
ES budget ($q = 95\%$)	TRP (historical)	Conditional TRP

Unconditional scenarios are the last 500 daily returns. Conditional scenarios are the same returns filtered by a univariate GJR-GARCH(1,1) per asset and rescaled to tomorrow's forecast volatility (multivariate FHS, Barone-Adesi et al., 1999). This keeps the historical cross-sectional dependence of the residuals and updates the scale. The ES effect is the row difference, holding the scenarios fixed; the conditioning effect is the column difference.

Execution and costs. Target weights are computed at each month end and become effective at the next open. Two execution rules are evaluated, both tracking actual holdings day by day: *daily rebalancing* back to target (a constant-mix strategy), and *monthly rebalancing with buy-and-hold*, in which weights drift with prices between rebalancing dates. On each trading day the amount traded is $\sum_i |w_{i,t}^{\text{target}} - w_{i,t^-}|$, where w_{t^-} are the drifted holdings, and a cost of $c \in \{0, 10, 25, 50\}$ basis points per unit traded is deducted from that day's return. Annual one-way turnover is half the annualised amount traded. Differences in excess Sharpe ratios (risk-free rate 2%) are assessed with a moving-block bootstrap (blocks of 20 days, 2,000 resamples; Künsch, 1989), which preserves volatility clustering; see Ledoit and Wolf (2008) for the same problem with HAC inference.

5.2 Results

Table 4 and Figure 2 summarise the backtest from July 2007 to October 2026.

Table 4: Tail risk parity decomposition, July 2007 to October 2026, monthly targets from 500-day windows. Turnover is realised annual one-way turnover. Excess Sharpe ratios (risk-free rate 2%) under monthly rebalancing with buy-and-hold, net of the cost per unit traded shown in the column header.

	Turnover		Excess Sharpe, cost per unit traded			
Strategy	Daily	Monthly	0 bp	10 bp	25 bp	50 bp
RP	53%	15%	0.93	0.92	0.91	0.89
TRP (historical)	54%	16%	0.97	0.96	0.95	0.92
Conditional RP	136%	95%	1.04	0.99	0.91	0.77
Conditional TRP	134%	95%	1.06	1.00	0.92	0.77
Differences in excess Sharpe, monthly buy-and-hold: estimate [95% block-bootstrap interval]						
ES effect (conditional scenarios)			+0.02 [-0.06, +0.09]	+0.01 [-0.06, +0.09]	+0.01 [-0.06, +0.08]	0.00 [-0.07, +0.07]
ES effect (unconditional)			+0.04 [-0.02, +0.11]	+0.04 [-0.03, +0.10]	+0.04 [-0.03, +0.10]	+0.03 [-0.03, +0.10]
Conditioning effect (volatility)			+0.11 [-0.07, +0.30]	+0.07 [-0.12, +0.25]	0.00 [-0.18, +0.18]	-0.12 [-0.30, +0.07]
Conditional TRP − RP			+0.13 [-0.08, +0.33]	+0.08 [-0.13, +0.29]	+0.01 [-0.20, +0.21]	-0.12 [-0.32, +0.09]

All figures below refer to monthly rebalancing with buy-and-hold; daily rebalancing gives the same conclusions with higher turnover.

Replacing volatility by ES adds almost nothing. Holding the scenarios fixed, the ES budget improves the excess Sharpe ratio by 0.04 with unconditional scenarios and by 0.02 with conditional scenarios before costs. Neither difference is significant (bootstrap 95% intervals $[-0.02, 0.11]$ and $[-0.06, 0.09]$). Because ES and volatility budgets have almost identical turnover, these differences hardly change with costs.

Conditioning on GARCH volatility is where the gross gain comes from, and costs remove it. Conditional RP improves on RP by 0.11 before costs (interval $[-0.07, 0.30]$), almost the whole gain of conditional TRP (0.13). But conditioning raises annual one-way turnover from 15% to 95% under monthly buy-and-hold, and from 53% to 136% under daily rebalancing. At 10 bp the gain falls to about 0.07, at 25 bp it is zero, and at 50 bp conditional strategies are about 0.12 *worse* than plain RP. The conclusion is the same under both execution rules; buy-and-hold has lower turnover and slightly higher net Sharpe ratios throughout.

Two further observations from the replication support this reading. In 2007–2012, the original AllianceBernstein sample, TRP is slightly worse than RP in these data; the gross gains come from after 2013. With a 99% ES level, historical TRP deteriorates (only five tail observations in a 500-day window).

6 Limitations

- **Markets and sample.** Six asset classes, mostly Chinese and US, 2005–2026 for the allocation and dependence tests. The conclusions are statements about this sample; tail behaviour in other markets or in periods without a 2008-type crisis may differ.
- **Asynchronous closes.** Weekly returns reduce but do not remove the bias from different closing times. Tail dependence between Chinese and Western assets may be understated.
- **Power.** Blocks of about 150 weekly observations contain roughly 15 decile exceedances per asset. Weak tail dependence, for example in the stock–bond pair, may go undetected. The panel null controls the size of the tests, not their power.

- **Scope of the methods.** Only return-based ES is tested. The option-implied ES used in the original tail-risk-parity proposal requires option data that do not exist for most Chinese assets. Static copulas are fitted per window; dynamic copulas (Patton, 2006) and regime-dependent models are not examined.
- **Costs and performance measure.** Costs are a constant number of basis points per unit traded, with no market impact and no differences between instruments. The excess Sharpe ratio is the primary criterion; the replication also reports tail Sharpe ratios and drawdowns, which lead to the same ranking.
- **Free data.** Index and ETF series come from free sources and were cross-checked (for example, two ETF sources agree to within 0.08% per day), but they are not vendor-grade.

7 Conclusion

Across the tests in this note, the information that changes risk forecasts and allocation outcomes is current volatility and correlation. Single-parameter tail-dependence coefficients reproduce correlation almost exactly. Tail dependence beyond correlation exists among risky assets but is invisible to the statistics used in practice. Expected-shortfall budgeting adds little to volatility budgeting, and the part of tail risk parity that does help before costs, conditioning on GARCH volatility, is consumed by its own turnover at realistic costs. In long ES backtests the commonly used Z_2 threshold is uninformative and should be replaced by a test with a proper null distribution. A practical implication is that tail-risk tools should be benchmarked against correlation-matched nulls and volatility-conditioned alternatives, net of costs, before they are credited with information of their own.

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